

Harnessing Frequency-Aware Network for Crack Detection in Road-Perceptive Autonomous Vehicles Scene

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Abstract—With the evolution of intelligent transportation systems, the integration of crack detection capabilities in vehicles has seen significant advancements. Existing deep-learning based crack detection methods, predominantly focused on the RGB domain, often fall short in complex scenarios where cracks blend seamlessly with their background. Recognizing that road cracks and surrounding pavement features exhibit greater discriminative potential in the frequency domain, we introduce a novel frequency-aware mechanism. Our frequency-aware network (FANet), operates in two step: a frequency-guided coarse crack localization step for initial crack identification, followed by a detail-preserving fine crack segmentation for enhanced precision. Leveraging different features from the backbone, FANet employs frequency-aware module for effective coarse positioning. Subsequently, a hybrid fusion module meticulously integrates high-level features via prior-guided corrections, further refining detection by incorporating shallow features for detailed correction of detected cracks. Empirical evaluations on five datasets demonstrate FANet’s superior performance.

Index Terms—Crack Detection, Frequency-aware Feature Learning, Road-perceptive Vehicles

I. INTRODUCTION

Maintaining the structural integrity of infrastructure is paramount, with crack detection playing a crucial role in safeguarding facilities against the detrimental effects of deterioration. Timely detection and remediation of cracks are essential for preventing potentially irreversible damage. The advent of autonomous vehicles and advanced environmental perception technologies [1]–[3] heralds a new era in automated road condition monitoring, with autonomous road-perceptive vehicles poised to revolutionize how we maintain our roadways [4]–[8]. Integrating crack detection capabilities into these vehicles offers a promising pathway for continuous road condition surveillance [9].

Traditional crack detection methods have primarily relied on digital image processing technologies, involving laborious procedures with complex preprocessing and initial crack identification, followed by refinement [17], [18]. These techniques, dependent on manually selected features and classical machine learning models, often falter when distinguishing cracks

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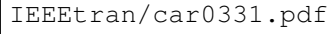
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Fig. 1. Performance of our proposed method and other methods, including AFFMM [10] and MSFE [11], in Crack500 [12], GAPs384 [13], Crack-Tree200 [14], CFD [15] and AEL [16].

against intricate backgrounds. The advent of deep learning marks a pivotal transition, ushering in a wave of sophisticated algorithms designed for specific types of cracks and conditions. This shift leverages the capabilities of convolutional neural networks (CNNs) to significantly improve road damage detection, utilizing hierarchical feature learning for edge detection [12], [19] and pioneering deep supervision and enhancement techniques for heightened performance. Moreover, multi-scale feature fusion has emerged to effectively marry local and global feature representations, with attention mechanisms fine-tuning feature selection for better discrimination [20]–[22].

Despite these advancements, deep learning methods predominantly focus on the RGB domain, searching for textural inconsistencies or ‘breakthrough points’ to pinpoint cracks. Yet, the nuanced characteristics of road surfaces frequently mask these defects, posing significant detection challenges.

Recently, several studies have introduced frequency-domain cues into image segmentation, such as FsaNet [23], LGFFM [24], FMamba [25], and FINet [26]. Although these methods validate the potential of frequency features, they typically treat frequency information as auxiliary input rather than achieving a deeply coupled frequency–spatial represen-



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Fig. 2. The road-perceptive vehicle may contain three steps. The first is to take photos of road condition. The second is to detection the crack by our proposed method. The third is to analysis the result and send back to vehicles.

tation. For instance, FsaNet focuses only on low-frequency tokens without adaptive fusion with spatial branches, LGFFM employs dual encoders with heavy computation, FMamba emphasizes global consistency while losing fine texture detail, and FINet is restricted to ultrasound imagery. These limitations reveal that existing frequency-based networks lack generalization and efficient integration with spatial representations, especially in complex natural scenes.

Addressing this, our study introduces the Frequency-aware network (FANet), a novel architecture that integrates RGB and frequency domain information. This dual-strategy approach aims to overcome the limitations of relying solely on RGB

information by enhancing the network's sensitivity to subtle textural features often obscured in the spatial domain. FANet represents a significant leap forward, offering enhanced accuracy in crack detection through a more holistic analysis of road surface characteristics. FANet commences its detection process with a frequency-guided coarse positioning phase, where the primary aim is to leverage frequency domain features for identifying potential crack locations, known as breakthrough points. Distinguished by a transformer backbone, FANet proficiently extracts different level features from the input. A specialized frequency-aware module (FAM) then distinguishes these features into high- and low-frequency

components, which correspond to the texture and contour details of the road surface, respectively. This division allows for a comprehensive representation of the frequency domain, enabling the coarse but accurate localization of road cracks. The process is enhanced by a neighbor interaction mechanism that promotes the integration of frequency-aware features across various levels, facilitating an initial yet refined crack detection.

Transitioning into the detail-preserving fine segmentation phase, FANet refines the previously identified crack locations through a process of prior-guided correction and cross-layer feature fusion. This refinement is achieved with the help of the hybrid fusion module (HFM), which manages the interplay of high-level features across different layers to boost the accuracy of the detection framework. The inclusion of shallow, high-resolution features ensures the meticulous delineation of structural information, leading to the production of the final crack detection output. This dual-phase approach, combining coarse positioning with fine localization, allows FANet to not only detect cracks with high precision but also to discern subtle features that traditional methods might overlook, setting a new standard in the field of road surface monitoring. In Fig. 1, our proposed framework demonstrates superior performance across all five datasets.

The core contributions of this paper are listed as follows:

- We propose a two-stage Frequency-Aware Network (FANet) that jointly models spatial and frequency information for robust crack detection. Unlike existing approaches that process frequency cues independently, FANet achieves unified frequency-spatial representation through frequency-guided coarse localization and detail-preserving fine segmentation, significantly enhancing precision and generalization under complex textures and noise.
- We design a fully integrated frequency-aware module capable of automatically distinguishing between low- and high-frequency features. This module significantly improves the localization of road cracks.
- We introduce a hybrid fusion module meticulously integrates high-level features via prior-guided corrections, further refining detection by incorporating shallow features for detailed correction of detected cracks. The performance in five different datasets demonstrates the effectiveness of our FANet.

II. RELATED WORK

A. Traditional Crack Detection Techniques

For traditional crack detection, Techniques such as intensity thresholding, morphological analysis, and classical machine learning strategies have been foundational, as evidenced by extensive different researches [27], [28]. A pioneering approach within this domain is the CrackTree algorithm, delineated in [14], which utilizes minimum spanning trees (MSTs) for the automated identification of cracks. Parallel to this, substantial efforts by researchers [16], [29], [30] have been devoted to leveraging minimal path techniques for crack region detection. Given the complex, irregular patterns of cracks, numerous

strategies have been formulated, emphasizing edge detection and bespoke feature engineering. An innovative methodology reported in [31] implements a dual-phase strategy, initiating with Region of Belief (RoB) seed extraction, succeeded by region-growing algorithms to coalesce disjointed regions for enhanced crack identification. Furthermore, the technique introduced in [15] makes use of random structure forests in crack detection, thereby facilitating a better comprehension of crack structures and diminishing noise artifacts. Nonetheless, these traditional methods, despite their utility in specific contexts, are often challenged by their susceptibility to noise and a heavy reliance on manual feature extraction, limiting their effectiveness in more intricate scenarios.

B. Advancements in Crack Detection via Deep Learning

The realm of crack detection has witnessed a transformative shift with the advent of deep learning technologies, heralding a new epoch of innovation and efficiency. This period is characterized by breakthroughs in deep learning that have revolutionized not only detection but also classification and segmentation within various fields [32]–[35]. Central to these advancements is the ability of deep learning models to autonomously discern and abstract pivotal, high-level features via an end-to-end learning process. **This capability has driven rapid progress in applying deep learning to crack detection tasks [36], [37].**

For example, an innovative hierarchical segmentation network that integrates FCNs with Conditional Random Fields (CRFs) and Guided Filtering (GF) is introduced in [19], aiming to exploit multi-layer features for meticulous refinement. Following this, encoder-decoder based FCNs have been developed, tailored to discriminate between crack and non-crack pixels with high precision [38]. The DeepCrackT model, as proposed in [39], enhances the encoder-decoder framework through a novel pairwise fusion technique of crack features at corresponding scales, significantly boosting detection capabilities. In pursuit of computational efficiency, the introduction of a lightweight crack detection network marks a significant milestone, incorporating split-exchange convolution and multi-scale feature fusion to streamline network operations and expedite detection processes [11].

However, relying solely on RGB information presents a significant limitation [40], as the spatial domain represented by RGB channels provides a direct yet occasionally superficial perspective of image content, potentially overlooking subtle features more discernible in the frequency domain. Recognizing this shortfall, our work introduces an innovative approach that incorporates frequency domain information into the crack detection process. By leveraging the frequency domain, we aim to reveal a wider spectrum of features, leading to more accurate and robust crack detection.

In addition to conventional RGB-based frameworks, several recent studies have explored incorporating frequency-domain cues into image segmentation. For instance, FsaNet [23] applies a discrete cosine transform (DCT) to enhance low-frequency representations, yet it lacks adaptive interaction between frequency and spatial branches, limiting its generalization to texture-rich or noisy environments. LGFFM [24]

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Fig. 3. Overall architecture of the proposed Frequency-Aware Network (FANet). The solid arrows denote feature flow between modules, while dashed lines represent skip connections used for cross-layer feature fusion. Each block (FAM, HFM, and Transformer-based backbone) corresponds to a specific processing stage described in Sections III-A–III-C.

introduces a dual-encoder structure combining CNN and Transformer backbones along with a Frequency Domain Mapping Module, which improves global perception but incurs high computational complexity and training instability. FMamba [25] embeds frequency enhancement into a lightweight Vision Mamba backbone, providing efficient feature extraction but prioritizing sequential consistency rather than fine-grained local texture cues. FINet [26] proposes a frequency-aware self-attention mechanism for ultrasound image segmentation, but it is task-specific and not easily generalizable to natural road scenes.

These methods collectively demonstrate the potential of frequency features in segmentation, yet they generally treat frequency cues as auxiliary enhancements rather than deeply integrating them with spatial representations. **In contrast, our proposed FANet establishes a unified two-stage frequency–spatial synergy, where the frequency-guided coarse localization phase captures global structure and the detail-preserving fine segmentation phase refines local details.** This design enables dynamic modulation between spectral and spatial features, achieving superior robustness, precision, and adaptability under complex road textures and illumination



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Fig. 4. The architecture of our proposed hybrid fusion module (HFM).

variations.

III. PROPOSED METHOD

A. Overview

Our objective is to harness and integrate the unique benefits of both the RGB and frequency domains, aiming to boost the discriminative capabilities for detecting road cracks via intelligent road-perceptive vehicles. In pursuit of this, we present the FANet, a novel architecture tailor-made for crack detection, as depicted in Fig. 3. FANet is architected around three core components: the Transformer-based feature extraction, the frequency-guided coarse crack localization step, and the detail-preserving fine crack segmentation step. Given an input image $S \in \mathbb{R}^{H \times W \times 3}$, FANet begins with the feature extraction with the Pyramid Vision Transformer (PVT) [41] as its feature extractor. This feature extractor generates multi-level feature maps $F_i, i \in \{1, 2, 3, 4\}$, each catering to different facets of the detection challenge. The first-level feature map F_1 , captures intricate details of the road surface essential for spotting minor cracks, whereas the subsequent levels (F_2, F_3, F_4) encapsulate progressively higher-level semantic information, pivotal for robust crack localization. In the frequency-guided coarse crack localization phase, our proposed novel frequency-aware module (FAM) extracts frequency domain features from the high-level features (F_2, F_3, F_4), which are then synergized using a tri-merge (T-M) decoder. This process yields a preliminary crack map S_1 .

After the coarse step, the detail-preserving fine crack segmentation step refines the initial map by using S_1 to steer the embedding of high-level features into a hybrid fusion module (HFM). This module facilitates layer-wise, prior-guided adjustments and fusion, significantly enhancing segmentation accuracy. After this step, the final output crack map is S_{output} .

B. Frequency-guided Crack Localization

Leveraging the concept that predators' hunting mechanisms often rely on frequency information to detect subtle anomalies in natural environments, we posit that similar advantages can be applied to road crack detection. This hypothesis is grounded

in observations that frequency domain methods, traditionally used in camouflaged object detection, can significantly enhance the detection of subtle road anomalies. Conventional techniques typically employ the DCT to transition from RGB to the frequency domain, a process that, despite its effectiveness, is hindered by the lack of flexibility due to offline processing requirements. However, recent advancements like the work [42] have introduced the use of octave convolutions, which enable the partitioning of image data into high and low-frequency components directly, thereby streamlining the detection workflow.

In Octave Convolution [42], feature maps are explicitly factorized into high- and low-frequency components along the channel dimension, represented as $\{F_i^h, F_i^l\}$. The high-frequency branch F_i^h captures rapidly varying fine-grained textures and edges, while the low-frequency branch F_i^l models smoothly varying contextual structures by operating at a lower spatial resolution (typically half). This separation enables the network to process detailed and structural information in a complementary way, providing an efficient approximation of frequency decomposition within the spatial domain.

These low-frequency components often represent smooth regions, such as uniform road surfaces, whereas high-frequency elements are indicative of potential cracks or damage due to their significant intensity variations. Inspired by the work [42], we explicitly factorize F_i and get the high- and low-frequency components $\{F_i^h, F_i^l\}$.

To further clarify how the proposed Frequency-Aware Module (FAM) facilitates crack localization, we provide a more detailed explanation of its mechanism. Specifically, FAM employs a learnable frequency decomposition to separate each feature map F_i into high- and low-frequency responses through adaptive filtering:

$$F_i^h = W_h * F_i, \quad F_i^l = W_l * F_i, \quad (1)$$

where W_h and W_l denote convolutional filters emphasizing fine-grained and smooth spectral components, respectively. This adaptive frequency gating enhances coarse localization by highlighting structural discontinuities while suppressing irrelevant texture noise, thus guiding the subsequent fine segmentation to focus on true crack contours.

After obtaining the high- and low-frequency information, the designed FAM dynamically integrates them through a dual-branch convolutional structure, where the high-frequency branch enhances texture-level crack cues and the low-frequency branch emphasizes global contour consistency. These two representations are then adaptively fused via learnable weights, allowing the network to emphasize salient crack details while suppressing irrelevant background noise. This mechanism enables FAM to achieve end-to-end frequency-aware feature interaction, forming a dual-band feature representation optimized for road crack detection. Intuitively, the high-frequency branch captures sharp discontinuities and fine textures that indicate crack edges, whereas the low-frequency branch maintains large-scale structural consistency. Their adaptive fusion lets the model highlight salient crack details while suppressing background clutter. Our proposed FAM achieves dynamic, end-to-end frequency information

processing, facilitating the direct learning of crack characteristics in real-time. This approach not only avoids the limitations associated with DCT but also benefits from the computational prowess of GPUs, allowing for efficient integration into diverse network structures. As shown in Fig. 3, the FAM process $f_i^* = \{f_i^h, f_i^l\}$ is mathematically represented as follows:

$$f_i^h = \text{Conv}(F_i^h) + \text{Upsample}(\text{Conv}(F_i^l), 2), \quad (2)$$

Equation (1) injects low-frequency contextual cues (after a 3×3 convolution and upsampling) into the high-frequency branch. This cross-branch message passing enriches boundary features with neighborhood context, making fine crack edges more reliable rather than over-fragmented.

$$f_i^l = \text{Conv}(F_i^l) + \text{Conv}(\text{Pool}(F_i^h, 2)), \quad (3)$$

Equation (2) performs the reverse interaction: pooled high-frequency responses are fed into the low-frequency branch. This stabilizes global shapes by attenuating spurious high-frequency noise while preserving meaningful discontinuities aligned with crack structures.

where $\text{Pool}(X, k)$ represents average pooling with a $k \times k$ kernel, and $\text{Upsample}(X, s)$ signifies upsampling by a scale factor of s through nearest neighbor interpolation.

Considering the significance of both high-frequency texture attributes for detailing surface irregularities and low-frequency contour attributes for outlining the broader features of the road surface, we integrate these components to form a comprehensive representation of frequency domain information tailored for road crack detection:

$$f_i = \{\text{Resize}(f_i^h) \odot W_i^h\} \oplus \{\text{Resize}(f_i^l) \odot W_i^l\}, \quad (4)$$

Finally, Equation (3) forms a frequency-balanced representation by resizing both branches to a common spatial dimension and applying learnable gates W_i^h and W_i^l for adaptive weighting. This process allows the network to dynamically control the contribution of each frequency component, producing features that jointly encode global contour integrity (from low-frequency cues) and precise edge localization (from high-frequency cues). Such a balanced representation strengthens both structural completeness and boundary sharpness for downstream crack detection.

where Resize denotes the operation to adjust features to a uniform dimension, W_i^h and W_i^l are learnable parameters, and $\oplus$ represents the element-wise addition. Here, $\odot$ denotes the element-wise multiplication.

The purpose of the frequency-guided coarse crack localization step is to initially locate the crack position. In order to fully utilize the effectiveness of the different features, we designed the T-M decoder so as to predict the crack location. This decoder leverages the semantic richness across layers via connections between adjacent layers, fostering a comprehensive feature representation as:

$$\begin{aligned} f'_3 &= \text{Conv}^\uparrow(f_4) \odot f_3, \\ f'_2 &= \text{Conv}^\uparrow(f_3) \odot f_2, \\ f'_{t-m} &= \text{Cat}(f_2, f'_2) \odot \text{Cat}(f_2, f'_3). \end{aligned} \quad (5)$$

Herein, $\text{Conv}^\uparrow$ represents upsampling followed by a 3×3 convolution, and $\text{cat}(A, B)$ denotes the concatenation of features A and B . The culmination of this phase is f'_{t-m} , leading to the formation of a coarse crack detection map S_1 , signifying the initial phase of road crack identification.

C. Detail-preserving Crack Segmentation

Upon the coarse localization via frequency-domain insights, we underscore the importance of refining these detections to ensure both completeness and precision. To this end, we introduce a detail-preserving segmentation strategy, crucial for the nuanced delineation of crack structural information. This approach enhances the integration of high-level features with prior corrections and channel associations, alongside the utilization of high-resolution features for accurate structural information mapping, as depicted in Fig. 3.

Central to this strategy is the HFM, shown in Fig. 4, a construct designed to seamlessly integrate features across layers, enriched by the insights from a coarse crack detection map $S_g = \{S_1, S_2\}$. To boost computational efficiency while retaining detection fidelity, input feature channels are initially condensed to 64, symbolized as F_i and F_{i+1} .

The HFM leverages the coarse crack detection map S_g as a spatial prior to refine subsequent layer features by modulating their activation intensity in crack-relevant regions. Specifically, S_g is multiplied with the feature maps F_{i+1} to generate a prior-weighted response that emphasizes potential crack areas while suppressing irrelevant background noise. This guided interaction enables the network to iteratively refine feature representations, achieving more accurate and stable crack localization across layers. This process is articulated through the multiplication of F_{i+1} with S_g , followed by upsampling to achieve refined feature outputs f_{i+1}^r :

$$f_{i+1}^r = \text{Upsample}(F_{i+1} \otimes S_g). \quad (6)$$

Given the richness of channel-specific cues in high-level features, the key to augmenting cross-layer feature dialogue and the module engages in channel-level association. This mechanism evaluates pixel-wise similarity across channels, employing a 3×3 convolution to construct a bottleneck, thus optimizing output channel dimensions:

$$A = \text{Conv}(F_i \odot (f_{i+1}^r)). \quad (7)$$

Subsequent steps involve deriving two weight maps, α and β , from the feature matrix A via dual 3×3 convolutions. These weights dynamically modulate F_i features, facilitating an advanced synthesis of cross-layer information:

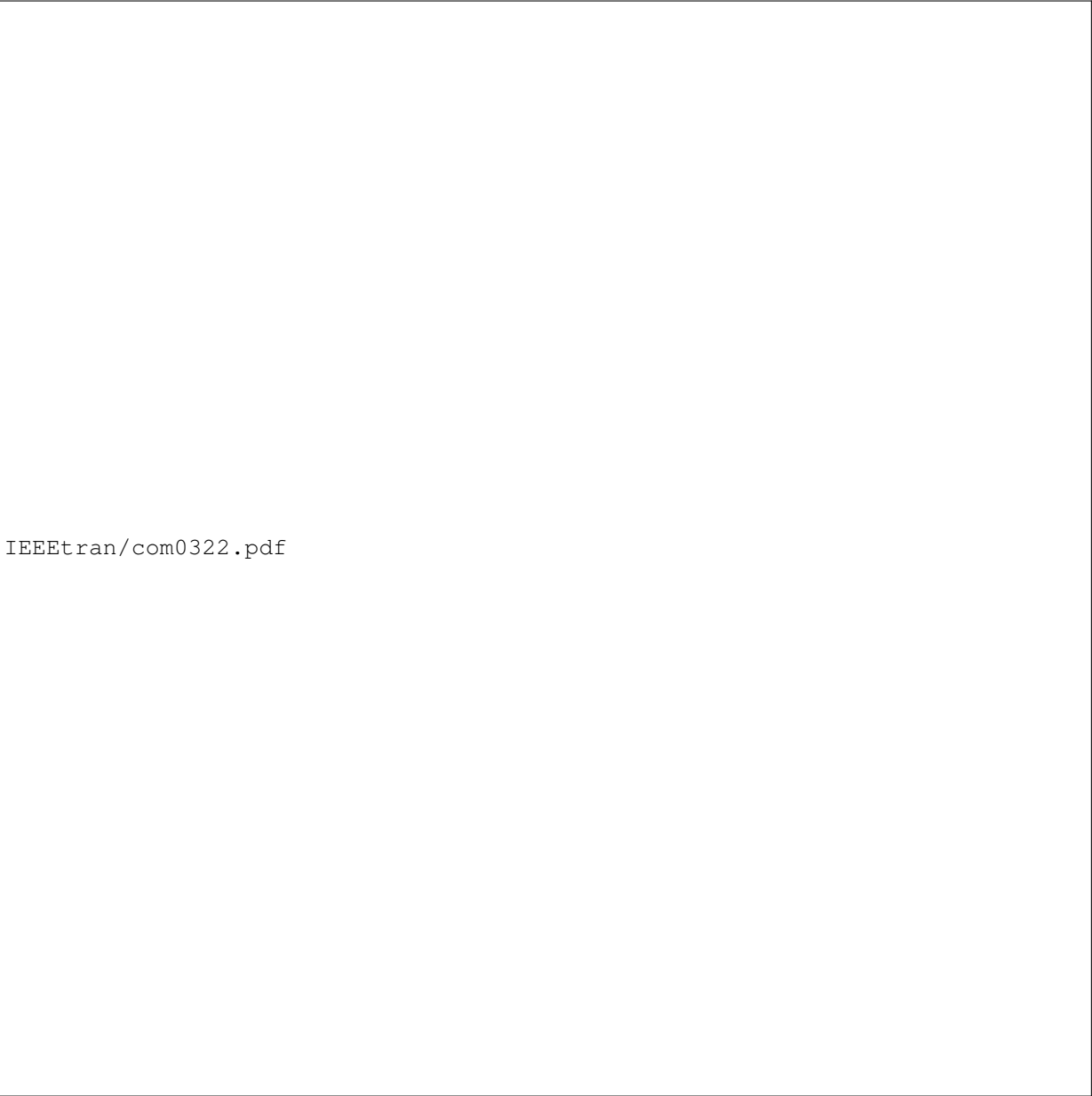
$$f_i^{\text{out}} = f_{i+1}^r + \text{Conv}(F_i) \times \alpha + \beta. \quad (8)$$

To complement these advanced techniques with detailed spatial insights, we use two 3×3 convolutions to generate the final output S_{output} .

Loss Function. In alignment with standard practices, we train the FANet through binary cross-entropy loss:

$$L = \text{BCE}(S_{\text{output}}, \mathcal{G}^T), \quad (9)$$

where $\mathcal{G}^T$ is the ground-truth.



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Fig. 5. Comparison between our FANet and other SOTA methods. It can be clearly observed that FANet achieves impressive performance in all these cases. Best viewed by zooming in.

IV. EXPERIMENTS

A. Implementation Details

FANet builds on the PVT in PyTorch pre-trained on ImageNet. We use the Adam optimizer, starting with a learning rate and weight decay of $1e - 4$. Input image are resized to 512×512 , and we train for 100 epochs on an NVIDIA 3090Ti GPU, with mini-batches of 4. Data augmentation strategies like random flipping and cropping are employed to improve robustness.

B. Datasets for Evaluation

Our model's efficacy is rigorously assessed using five widely recognized datasets, each offering unique challenges that simulate the complexities of crack detection scenarios.

Crack500 [12]: A comprehensive dataset tailored for crack detection, Crack500 consists of 500 images, partitioned into 250 for training, 50 for validation, and 200 for testing. It encompasses a diverse array of crack conditions, including but not limited to occlusions, variable contrast, shadow effects, and ambient noise, accurately mirroring the intricate nature of real-life crack detection challenges.

GAPs384 [13]: Designed to tackle the intricacies of extreme lighting conditions on road surfaces, GAPs384 categorizes cracks into two distinct categories based on lighting: excessively dark and overly bright areas. This categorization underlines the challenge of distinguishing between cracks under varying lighting conditions. The dataset is comprised of 384 images, meticulously annotated to facilitate detailed analysis, with image dimensions of either 640×540 or 440×540 .

CFD [15]: Specializing in cracks on concrete surfaces, the CFD dataset includes 118 images, each sized at 480×320 . The dataset is characterized by its portrayal of common real-world disturbances such as uneven lighting, shadows, obstructions, and moisture accumulation, thus providing a robust platform for evaluating crack detection algorithms.

Cracktree200 [14]: Comprising 206 images of pavement cracks at a resolution of 800×600 , Cracktree200 emphasizes the detection of slender crack formations. The dataset is notable for its representation of fine cracks amidst complex backgrounds, including tree shadows, ambiguous road markings, and the intersection of cracks with such road features.

AEL [16]: The AEL dataset amalgamates three subsets: Aigle-RN, ESAR, and LCMS. Aigle-RN includes 38 pixel-annotated images for the assessment of French pavements, whereas ESAR and LCMS offer 15 and 5 images, respectively, without specified lighting conditions, further diversifying the dataset's utility in crack detection research.

C. Evaluation Metrics Explained

In our evaluation, we report four representative metrics to comprehensively assess the performance of our network: the F1 score, Average Intersection over Union (AIU), Average Precision (AP), and Frames Per Second (FPS).

F1 (Dice Score). The F1 score, also known as the Dice Score in binary segmentation tasks, is the harmonic mean of precision (P) and recall (R):

$$F1 = 2 \cdot \frac{P \cdot R}{P + R}. \quad (10)$$

It jointly evaluates the accuracy of the predicted cracks (precision) and the completeness of detection (recall), thereby providing a balanced measure of detection performance.

AIU (IoU). The Average Intersection over Union (AIU) measures the mean overlap ratio between predicted and ground-truth crack regions, averaged over all thresholds. Since AIU is the averaged form of IoU, it is mathematically equivalent to the conventional IoU metric but provides a more stable representation under varying detection thresholds.

AP. Average Precision (AP) summarizes the precision-recall relationship into a single scalar value, reflecting the model's overall discriminative ability to separate cracks from the background.

FPS. Frames Per Second (FPS) quantifies inference efficiency by measuring the number of test images processed per second on an NVIDIA 3090 Ti GPU. Higher FPS values indicate faster model inference and better real-time potential.

Since AIU and F1 correspond to IoU and Dice Score respectively, we avoid redundant metrics and instead include FPS to complement segmentation accuracy with computational efficiency analysis.

D. Comparisons with the State-of-the-Art

In our evaluation, FANet's performance is rigorously benchmarked against a wide spectrum of state-of-the-art deep learning models in the domain of road crack detection. To ensure

equitable comparison parameters, all considered methods are trained on the same dataset under a uniform learning rate setting. Moreover, if authors provide their results, we directly use these crack maps. The comparative analysis encompasses a diverse array of methodologies. The comparison methods include BASNet [43], SegNet [44], RCF [45], FPHBN [12], FCN [46], D-Crack [39], UHDN [47], D-CrackM [19], FFEDN [48], CrackF-r [49], MSFE [11], AMMFF [10], CarNet [50], HFF [51], FsaNet [23], LGFFM [24], and FMamba [25] as our primary competitors. The latter group, comprising FFEDN, AMMFF, HFF, MSFE, CrackFormer and CarNet, represents the forefront of advancements published from 2022 to 2024. These methodologies underscore the rapid evolution within the field, each bringing novel perspectives and solutions to the challenges inherent in crack detection using deep learning frameworks.

In addition, we also include three recent frequency-domain segmentation networks for a fair comparison, namely FsaNet [23], LGFFM [24], and FMamba [25]. These models extend segmentation architectures into the frequency domain and provide valuable insights into how spectral representations can enhance visual understanding. Although FsaNet, LGFFM, and FMamba all incorporate frequency cues, they mainly treat frequency information as an auxiliary enhancement rather than a deeply integrated representation. Specifically, FsaNet applies a discrete cosine transform (DCT) to low-frequency tokens but lacks adaptive interaction between frequency and spatial branches, leading to limited generalization under texture-rich or noisy outdoor scenes. LGFFM introduces a dual-encoder architecture combining CNN and Transformer features in the frequency domain, which enhances global perception yet suffers from high computational complexity and training instability. FMamba further integrates frequency enhancement into a lightweight Vision Mamba backbone; however, its state-space modeling emphasizes sequential consistency, which is less effective for fine-grained cracks with high-frequency fluctuations.

By contrast, our proposed FANet establishes a *two-stage frequency-spatial synergy*, consisting of a frequency-guided coarse localization followed by a detail-preserving fine segmentation. This design enables dynamic modulation between high- and low-frequency representations under spatial constraints, yielding superior precision, robustness, and computational efficiency across diverse datasets.

E. Quantitative Results

Table. I highlights our method's outstanding performance on five datasets through F1, AIU, and AP metrics. It significantly outshines existing state-of-the-art techniques, marking a new era in crack detection efficiency. Specifically, on the Crack500 dataset, our approach not only leads with an F1 score of 0.726 but also showcases superior AIU and AP scores of 0.563 and 0.572, respectively. This performance trend is mirrored across datasets such as CFD, CrackTree, AEL, and GAPS384, emphasizing our method's adaptability and robustness.

Such consistent leading scores across diverse datasets signal our method's advanced capability in enhancing road safety

TABLE I

COMPARISON OF OUR PROPOSED METHOD’S PERFORMANCE WITH OTHER METHODS USING F1, AIU, AND AP METRICS ACROSS FIVE DATASETS. HIGHER VALUES IN F1, AIU, AND AP INDICATE SUPERIOR PERFORMANCE. VALUES HIGHLIGHTED IN RED INDICATE THE BEST PERFORMANCE, WHILE VALUES IN GREEN REPRESENT THE SECOND-BEST RESULTS FOR EACH DATASET.

Metric		Method																	
		BASNet	FPHBN	SegNet	FCN	D-Crack	UHDN	D-CrackM	UNet	FFEDN	CrackF-r	MSFE	AMMFF	HFF	CarNet	FsaNet	LGFFM	FMamba	Ours
FPS		7.9	9.3	8.6	8.8	6.9	8.6	11.4	9.3	9.4	10.5	11.9	10.5	8.8	22.2	25.6	16.2	11.3	10.2
Crack500	AIU	0.508	0.492	0.500	0.383	0.505	0.471	0.440	0.443	0.492	0.498	0.495	0.507	0.494	0.503	0.535	0.527	0.522	0.563
	AP	0.510	0.498	0.510	0.385	0.510	0.473	0.444	0.449	0.501	0.505	0.496	0.515	0.501	0.511	0.545	0.538	0.532	0.572
	F1	0.689	0.681	0.649	0.529	0.678	0.684	0.659	0.678	0.624	0.640	0.682	0.683	0.620	0.642	0.695	0.684	0.676	0.726
CFD	AIU	0.163	0.174	0.162	0.022	0.115	0.206	0.083	0.230	0.390	0.279	0.356	0.255	0.334	0.347	0.372	0.365	0.360	0.391
	AP	0.173	0.180	0.163	0.022	0.122	0.213	0.091	0.233	0.394	0.281	0.363	0.265	0.339	0.351	0.381	0.375	0.369	0.397
	F1	0.275	0.379	0.261	0.234	0.414	0.569	0.341	0.550	0.595	0.610	0.599	0.616	0.630	0.611	0.618	0.607	0.600	0.644
CrackTree	AIU	0.076	0.044	0.067	0.008	0.027	0.048	0.024	0.072	0.136	0.158	0.089	0.099	0.148	0.160	0.236	0.230	0.226	0.252
	AP	0.082	0.045	0.071	0.012	0.031	0.054	0.033	0.080	0.145	0.168	0.094	0.106	0.149	0.169	0.240	0.235	0.231	0.253
	F1	0.190	0.137	0.188	0.109	0.120	0.190	0.115	0.237	0.267	0.224	0.263	0.290	0.315	0.253	0.341	0.333	0.328	0.362
AEL	AIU	0.155	0.079	0.131	0.026	0.064	0.108	0.038	0.112	0.270	0.178	0.192	0.299	0.272	0.232	0.297	0.292	0.288	0.312
	AP	0.156	0.085	0.139	0.036	0.069	0.111	0.048	0.114	0.277	0.185	0.200	0.301	0.275	0.242	0.302	0.298	0.294	0.317
	F1	0.309	0.020	0.312	0.230	0.250	0.359	0.242	0.404	0.430	0.442	0.436	0.460	0.472	0.483	0.481	0.473	0.468	0.505
GAPS384	AIU	0.053	0.081	0.060	0.021	0.066	0.092	0.047	0.077	0.170	0.232	0.096	0.159	0.177	0.213	0.252	0.247	0.244	0.266
	AP	0.057	0.087	0.064	0.026	0.076	0.095	0.047	0.078	0.178	0.238	0.100	0.164	0.179	0.218	0.251	0.247	0.243	0.265
	F1	0.136	0.187	0.137	0.112	0.241	0.333	0.268	0.259	0.389	0.379	0.497	0.383	0.435	0.394	0.497	0.489	0.483	0.522

and maintenance through improved crack detection. This underscores the significant leap our approach represents in the application of computer vision in infrastructure care.

Analysis of Inference Efficiency. We conduct a comprehensive comparison of the inference efficiency of our FANet and other crack detection methods in Table I. All results are measured on an Intel(R) Xeon(R) Gold 5218R CPU and an NVIDIA RTX 3090 GPU. The Frames-Per-Second (FPS) values are calculated as the reciprocal of the average inference time per image. As shown in Table I, our FANet achieves a balanced trade-off between accuracy and efficiency — attaining competitive FPS while delivering significantly higher segmentation accuracy than other baselines. This demonstrates that the proposed frequency-spatial synergy design enhances detection robustness without introducing notable computational overhead.

F. Qualitative Results

Fig. 5 showcases the proficiency of our FANet in detecting various types of road cracks, juxtaposed against the outcomes from competing methods. The visual comparisons strikingly illustrate the superior segmentation accuracy of FANet, highlighting its capability to delineate a wide array of crack categories with remarkable precision. This adeptness in accurately identifying and segmenting different forms of cracks underscores the method’s robustness and its nuanced understanding of the crack detection task. Such precision is indispensable in fields where accurate crack detection plays a pivotal role, including structural health monitoring, quality assurance in manufacturing processes, and geological assessments. Through these visualized outcomes, the efficacy and applicational potential of FANet in critical real-world scenarios are vividly demonstrated, affirming its value as a significant advancement in the domain of crack detection.

G. Ablation Studies

To elucidate the contributions of distinct components within FANet towards enhancing road crack detection, we conducted a series of ablation studies. These studies dissect FANet

into its constituent elements, namely, the frequency-aware module (FAM), the tri-merge (T-M) decoder, and the hybrid fusion module (HFM), with the PVT backbone serving as the ‘Baseline’ for comparison. This baseline represents our initial approach to detecting road cracks without the advanced mechanisms introduced by FANet. The outcomes of these comparative analyses are detailed in Table II, demonstrating the incremental improvements in crack detection performance attributable to each module. In particular, we further divide the Frequency-aware Module (FAM) into three configurations, namely Low-Frequency, High-Frequency, and Full fusion, to quantify their individual and joint contributions. The results reveal that both low- and high-frequency cues independently enhance detection accuracy, while their combination achieves optimal balance between precision and stability. To better assess computational efficiency, we also report inference speed (FPS) for each configuration, reflecting the module-wise contribution to overall runtime. Notably, adding frequency and fusion modules leads to consistent performance gains with only marginal computational cost, highlighting the lightweight design of FANet. As illustrated in Fig. 6, these findings are visually consistent with the observed improvements from high- and low-frequency cues.

Effectiveness of FAM. Our FAM efficiently reduce spatial redundancy while autonomously learning high- and low-frequency features. As indicated in Table. II, incorporating FAM into the baseline (*i.e.*, Baseline+FAM) results in significant performance improvements across all metrics when compared to the PVT backbone alone. This enhancement is primarily due to FAM’s capacity to capture rich frequency-aware information, particularly high-frequency details crucial for the initial detection of road cracks. Another key advantage of the FAM is its ability to learn in an online manner without the need for additional offline processes, enhancing its applicability in real-world scenarios. The flexibility and effectiveness of the FAM render it a potent tool for accurately detecting road cracks, especially by delineating subtle structural information that may blend into the surrounding pavement. Moreover, FAM’s compatibility with various frameworks underscores its potential to further refine crack detection capabilities by

TABLE II

ABLATION STUDIES OF THE PROPOSED FANet ON THREE DATASETS (GAPS384, Crack500, AND CFD). WE REPORT AIU, AP, F_1 , AND FPS TO ANALYZE THE IMPACT OF EACH MODULE AND THE CONTRIBUTION OF FREQUENCY COMPONENTS (LOW-FREQ, HIGH-FREQ, AND FULL) IN FAM. THE RESULTS DEMONSTRATE THAT BOTH HIGH- AND LOW-FREQUENCY CUES IMPROVE PERFORMANCE, WHILE THE INTEGRATION OF HFM ACHIEVES THE BEST BALANCE BETWEEN ACCURACY AND EFFICIENCY.

Models	GAPS384			Crack500			CFD			FPS
	AIU	AP	F_1	AIU	AP	F_1	AIU	AP	F_1	
Baseline	0.166	0.164	0.422	0.461	0.475	0.628	0.291	0.300	0.543	13.2
Baseline + FAM (Low-Freq)	0.192	0.190	0.450	0.538	0.545	0.708	0.319	0.325	0.570	11.3
Baseline + FAM (High-Freq)	0.195	0.193	0.448	0.541	0.549	0.706	0.322	0.327	0.575	11.2
Baseline + FAM (Full)	0.200	0.198	0.458	0.550	0.556	0.711	0.324	0.330	0.578	10.8
Baseline + FAM + T-M	0.231	0.233	0.491	0.528	0.537	0.694	0.358	0.363	0.612	10.4
Baseline + FAM + T-M + HFM	0.266	0.265	0.522	0.563	0.572	0.726	0.391	0.397	0.644	10.2

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Fig. 6. Comparison of different settings output from FAM. (a) Input. (b) Ground-truth. (c) means the prediction using the frequency fusing mechanism. (d) means the prediction only using high-frequency information. (e) means the prediction only using low-frequency information.

highlighting the nuanced features of road surfaces.

Effectiveness of T-M. The purpose of our proposed T-M is to fuse the frequency features of several different layers to get a better crack localization map, which can better guide the next step. Our ablation study demonstrated that decoding with features from multiple layers significantly enhances performance compared to using a single layer. This improvement can be attributed to the rich, multi-scale information obtained from diverse layers, which encompasses both detailed textures and abstract semantics. Integrating these features provides a more comprehensive representation, improving the accuracy and robustness of the crack localization map.

Effectiveness of HFM. The initial coarse crack detection achieved in the first step could be further optimized. To address this, we introduce the HFM aimed at maximizing the utility of the coarse detection map alongside neighboring features. The implementation of HFM within the detail-preserving crack segmentation step is evidenced by the improved metrics in the last row of Table. II. Incorporating HFM not only elevates the

overall performance, particularly in AIU, AP, and F_1 scores, but also underscores the module's pivotal role in advancing road crack detection capabilities.

The HFM's efficacy stems from its comprehensive approach to integrating the initial coarse prediction mask with layer-wise feature interactions. Initially, the coarse detection mask offers a precise delineation of potential crack areas, facilitating the extraction of relevant features. Subsequent channel-wise correlation of adjacent layers further refines the feature set, enhancing crack representation for more accurate detection. This dynamic interaction, enabled by HFM's learning of channel correlations to generate adaptable weight maps and refine features, proves more effective than straightforward concatenation methods. Such improvements highlight the value of fusing prior detection insights with cross-layer feature interactions, demonstrating HFM's crucial contribution to the nuanced process of road crack detection refinement.

In summary, the frequency-guided crack localization step in FANet initially delineates critical areas for potential road cracks, leveraging hierarchical frequency-aware semantic information to highlight these regions. The subsequent detail-preserving crack segmentation step further refines this detection by effectively differentiating the cracks from their surrounding complex background. This refinement is achieved through the integration of high-resolution details, correlation features between adjacent layers, and the initial coarse detection map. Collectively, these stages enable FANet to provide a precise and robust solution for the nuanced task of road crack detection, demonstrating its capacity to navigate the challenges posed by subtle and complex crack patterns within diverse road surfaces.

V. CONCLUSION

In this study, we introduce the FANet, a novel approach tailored to overcome the intricacies of road crack detection through the synergistic use of RGB and frequency domains. At the heart of FANet lies the frequency-aware module, designed to meticulously segregate frequency information. This module facilitates the generation of an initial coarse mask, effectively highlighting potential crack areas in the early stages of detection. Further refinement is achieved via a detail-preserving crack segmentation step, augmented by a hybrid fusion module, to enhance the accuracy of the

coarse prediction map. Our extensive evaluation, encompassing comprehensive comparisons and ablation studies three crack detection datasets, has underscored FANet's capability and efficiency. The results not only demonstrate FANet's superiority in identifying road cracks but also lay the groundwork for future research. This contribution is poised to inspire further advancements in leveraging frequency domain insights for solving complex challenges within the multimedia and infrastructure maintenance domains.

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